

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 February 2026
Team ID	699d5f82fdf409fdbd1bec66
Project Name	Malaria Detection using Deep Learning
Maximum Marks	4 Marks

Technical Architecture:

The proposed system is an AI-based malaria detection platform that analyzes microscopic blood smear images and automatically classifies them as Parasitized or Uninfected using deep learning models.

The system consists of multiple layers:

1. User Interface Layer
 - Provides a web-based interface where users can upload blood smear images.
2. Application Logic Layer
 - Handles image preprocessing, model loading, and prediction logic.
3. Machine Learning Layer
 - Deep learning models trained on malaria cell image datasets perform classification.
4. Data Layer
 - Stores datasets and trained model files.
5. Deployment Layer
 - The system runs on a local server or cloud environment and provides real-time predictions.

The architecture integrates data preprocessing, deep learning model training, and web-based deployment to create an end-to-end automated diagnostic system.

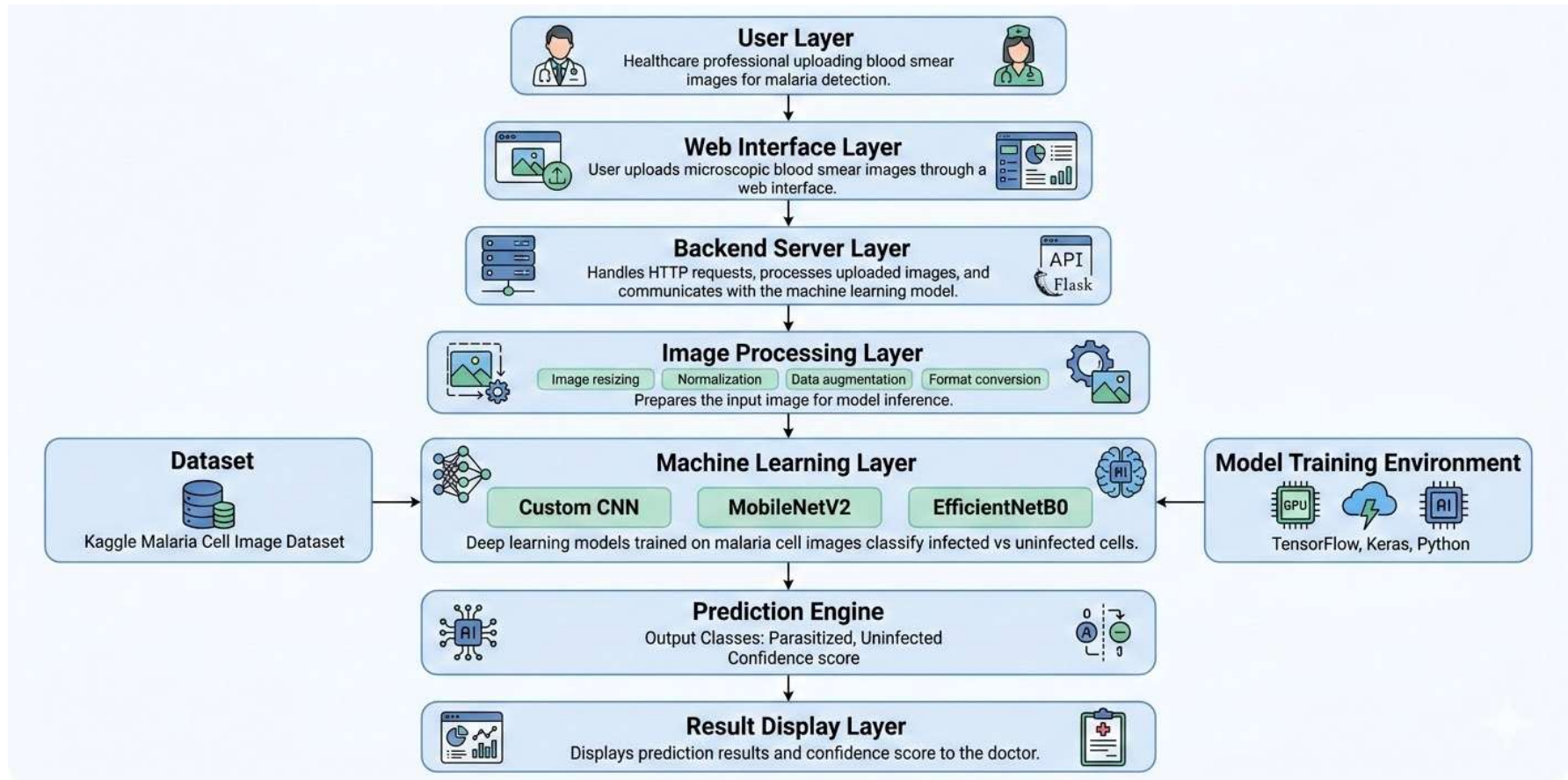


Table-1 : Components & Technologies:

	Component	Description	Technology
1.	User Interface	Interface through which users upload blood smear images and view predictions	HTML, CSS, JavaScript/ React.js

2.	Application Logic-1	Handles image upload, preprocessing, and request handling	Python
3.	Application Logic-2	Model inference logic for predicting infected or uninfected cells	TensorFlow, Keras
4.	Application Logic-3	API endpoints for prediction requests and responses	Flask Web Framework
5.	Database	Dataset used for training and validation of models.	Kaggle Malaria Dataset
6.	File Storage	Stores trained model files and uploaded images	Local File System
7.	External API-1	Kaggle API used for downloading malaria image dataset	Kaggle API
8.	Machine Learning Model	Detects malaria parasites in blood smear images	Custom CNN, MobileNetV2, EfficientNetB0
9.	Infrastructure (Server / Cloud)	Deployment environment for running the web application	Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The system uses open-source frameworks for deep learning, data processing, and web development	TensorFlow, Keras, Flask, NumPy, Pandas
2.	Security Implementations	Ensures secure data handling and access control for uploaded medical images	HTTPS, Secure File Handling, Authentication
3.	Scalable Architecture	The architecture supports scaling by deploying models on cloud servers and handling multiple image requests simultaneously	Cloud Computing, Containerization
4.	Availability	The system can be deployed on cloud platforms with high availability and uptime	AWS, Google Cloud, Load Balancers